## 1. Core Concepts & Modelling

***LTI state-space:***

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu + D, \quad y(t) = C\mathbf{x}$$

$A$=dynamics (main), $B$=input, $C$=output, $D$=disturbance. $A\in\mathbb{R}^{n\times n}$, $B\in\mathbb{R}^{n\times 1}$, $C\in\mathbb{R}^{1\times n}$. Feedthrough form: $y=C\mathbf{x}+Du$.

***ODE → SS:*** for $\ddot{x} + a\dot{x} + bx = u$, $\mathbf{x} = (x, \dot{x})^T$:

$$A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}, \; B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \; C = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

3rd order $\dddot{x} + a\ddot{x} + b\dot{x} + cx = u$, $\mathbf{x} = (x, \dot{x}, \ddot{x})^T$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -c & -b & -a \end{pmatrix}, \; B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \; C = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

***TF → SS:*** for $G(s) = \dfrac{b_0}{s^2 + a_1 s + a_0}$:

$$A = \begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix}, \; B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \; C = \begin{pmatrix} b_0 & 0 \end{pmatrix}$$

3rd order $G(s) = \dfrac{b_0}{s^3 + a_2 s^2 + a_1 s + a_0}$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{pmatrix}, \; B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \; C = \begin{pmatrix} b_0 & 0 & 0 \end{pmatrix}$$

***Mass-spring-damper:*** $m\ddot{y} + c\dot{y} + ky = u$, $G(s) = \dfrac{1}{ms^2 + cs + k}$.

## 2. Linearisation (Lec 2)

For $\dot{x} = f(x, u)$:
**(1)** Equilibrium $f(x_e, u_e) = 0$.
**(2)** Taylor (1st order):

$$\dot{x} \approx f(x_e, u_e) + \frac{\partial f}{\partial x}\Big|_e (x - x_e) + \frac{\partial f}{\partial u}\Big|_e (u - u_e)$$

General (scalar) form: $F(v_0+\delta v) \approx F(v_0) + \dfrac{\partial F}{\partial v}\Big|_{v_0} (v - v_0)$.

**(3)** Deviation vars $\delta x = x - x_e$, $\delta u = u - u_e$:

$$\dot{\delta x} \approx \frac{\partial f}{\partial x}\Big|_e \delta x + \frac{\partial f}{\partial u}\Big|_e \delta u$$

Multivariable: $A = \frac{\partial f}{\partial \mathbf{x}}\big|_e$ (Jacobian), $B = \frac{\partial f}{\partial u}\big|_e$.

## 3. First-Order Systems (Lec 3)

$\dot{x} + ax = u$, free resp.: $x(t) = e^{-at}x(0)$.
**Stability**: pole at $s= -a$. $a>0 \Rightarrow$ decays, **stable**; $a<0 \Rightarrow$ grows, **unstable**.
Step ($u$=const), $x(0)$=$x_0$:

$$x(t) = \left(x_0 - \tfrac{u}{a}\right)e^{-at} + \tfrac{u}{a}$$

Steady state: $x(\infty) = u/a$. Time constant $\tau = 1/a$.
***Time specs:*** rise $t_r \approx \frac{2.2}{a}$=$2.2\tau$ (10–90%); settle $t_s \approx \frac{3.91}{a} \approx \frac{4}{a}$=$4\tau$ (2%).
***Transfer function:*** $G(s) = \dfrac{\text{output}}{\text{input}} = \dfrac{Y(s)}{U(s)}$, $y = G(s)u$, $u = Ue^{st}$.
$G(s) = \dfrac{X(s)}{U(s)}$. For $\ddot{x} + 3\dot{x} + 2x = u$: $G = \frac{1}{s^2+3s+2}$.
***DC gain → final value:*** for *any stable* $G$, step of size $U$:

$$x(\infty) = G(0)\cdot U \quad \text{(set all derivs = 0)}.$$

$G(0)$ sign = direction ($G(0)<0 \Rightarrow$ output dips below 0); magnitude = final value. Generalises $x_\infty$=$u/a$.

## 4. Proportional / Steady-State Error

Plant $\dot{x} + ax = u$, $u = K_P(r - x)$:

$$\dot{x} + (a{+}K_P)x = K_P r$$

$x(\infty) = \frac{K_P r}{K_P+a}$, error $e(\infty) = r\,\frac{a}{K_P+a}$.
With disturbance $d$, noise $n$:

$$x_{ss} = \frac{K_P(r-n) + d}{a + K_P}$$

Large $K_P \Rightarrow$ small error (**feedback stabilisation**: pick $K_P > -a$). **Integral** action $\Rightarrow$ zero steady-state error.
***SS error (unity fb, unit step):*** $e(\infty) = \dfrac{1}{1+G(0)}$ ($G(0)$=DC loop gain; position const $K_p$=$G(0)$).
***System type:*** # poles of $L$=$CG$ at $s$=0: $0, 1, 2 \Rightarrow$ Type $0, 1, 2$.

| | Type 0 | Type 1 | Type 2 |
|---|---|---|---|
| Step | $\dfrac{1}{1+K_p}$ | 0 | 0 |
| Ramp | $\infty$ | $\dfrac{1}{K_v}$ | 0 |
| Parab. | $\infty$ | $\infty$ | $\dfrac{1}{K_a}$ |

## 5. Homogeneous Solns & Poles (Lec 4)

Char. eq. of $\ddot{x} + a\dot{x} + bx = 0$: $\lambda^2 + a\lambda + b = 0$.
***Real distinct:*** ($a^2{-}4b{>}0$): $x_h = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$.
***Real repeated:*** ($a^2{-}4b{=}0$): $x_h = (C_1 + C_2 t)e^{\lambda^* t}$.
***Complex:*** $\lambda = -\sigma \pm j\omega_d$:

$$x_h = e^{-\sigma t}(C_3 \cos\omega_d t + C_4 \sin\omega_d t)$$

**Stability**: all poles (roots of denom.) have negative real part $\Leftrightarrow x(t) \to 0$ for any $x(0)$, zero input.
**Necessary cond.**: char. poly has *all coefficients present & positive* (same sign) — any missing/sign change $\Rightarrow$ unstable. Real, distinct, negative poles $\Rightarrow$ stable & non-oscillatory.
$G(s) = \dfrac{b_1 s + b_0}{s^2 + a_1 s + a_0}$; **poles**=denom roots, **zeros**=numer roots.

## 6. Second-Order Systems (Lec 5)

Standard: $G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.
Poles $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$.

$$\sigma = \zeta\omega_n, \quad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2}, \quad \zeta = \sigma/\omega_n$$

$\zeta$ = sine of angle from imag. axis. Underdamped: $0 < \zeta < 1$.
***Envelope:*** $x(t) = e^{-\sigma t}\sin(\omega_d t)$.
**Response specs:** $M_p$=$e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ (overshoot), $t_p$=$\dfrac{\pi}{\omega_d}$ (peak time), $t_s$≈$\dfrac{4}{\zeta\omega_n}$ (2% settle), $t_r$≈$\dfrac{1.8}{\omega_n}$ (rise 10–90%).

## 7. PID Control (Lec 5)

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}$$

- **P**: fast, may leave SS error / oscillate.
- **PI**: removes SS error, slower transient.
- **PD**: adds damping, faster, noise-sensitive, adds a zero.
- **PID**: removes SS error + improves transient.

$K_p{\uparrow} \Rightarrow \omega_n{\uparrow}$ (faster), $\zeta{\downarrow}$ (more overshoot).
***Rate feedback:*** $u = K_P(r - x) - K_D\dot{x}$ (avoids derivative kick):

$$G(s) = \frac{K_P + K_D s}{s^2 + (1+K_D)s + K_P}$$

zero at $s = -K_P/K_D$.
**PD design ex.** $\ddot{x} + 3\dot{x} + 2x = u$, $u = K_p(r - x) - K_d\dot{x}$:

$$G = \frac{K_p}{s^2 + (3+K_d)s + (K_p + 2)}$$

match to $s^2 + 2\zeta\omega_n s + \omega_n^2$.

## 9. State-Space Response (Lec 6)

***Matrix exponential:*** $e^{At} = \displaystyle\sum_{k=0}^{\infty}\frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \cdots$

Free resp.: $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$.
Forced: $\mathbf{x}(t) = e^{At}\mathbf{x}_0 + \displaystyle\int_0^t e^{A(t-\tau)}Bu\,d\tau$.
***TF from SS:*** $G(s) = C(sI - A)^{-1}B + D$.
Output to $u = Ue^{st}$: $y = Ce^{At}\mathbf{x}_0 + G(s)Ue^{st}$.
**Stable** $\Leftrightarrow$ all eigenvalues of $A$ have Re $< 0$.
$e^{At} = Te^{Jt}T^{-1}$ (Jordan); repeated eig give $t^k$ terms.

## 10. Reachability / Control (Lec 7)

***Reachability matrix:*** (3rd order: $3\times 3$)

$$W = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

Reachable $\Leftrightarrow \det W \neq 0$.
Equivalent: can place *any* closed-loop poles via $K$.
***State feedback:*** $u = -K\mathbf{x}$, closed loop $\dot{\mathbf{x}} = (A - BK)\mathbf{x}$.
Design: $\det(sI - (A - BK)) \overset{!}{=}$ desired char. poly, **match coefficients**.
***Transformation:*** $T = W\tilde{W}^{-1}$ ($\tilde{W}$ from reachable canonical form), $K = \tilde{K}T$.
***Reachable canonical form:*** for $\ddot{y}{+}a_1\dot{y}{+}a_2 y{=}u$ (top-row companion):

$$\tilde{A} = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix}, \; \tilde{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

char. poly $s^2{+}a_1 s{+}a_2$ read off top row.
***Dominant poles:*** choose 2nd-order pair for transient; place rest 5–10× further left, low overshoot.

## 11. Observability / Observers (Lec 8)

***Observability matrix:*** (3rd order: $3\times 3$)

$$W_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Observable $\Leftrightarrow \det W_o \neq 0$: recover $\mathbf{x}(0)$ from $u, y$ on $[0, T]$.
***Luenberger observer:***

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + Bu + L(y - C\hat{\mathbf{x}})$$

Error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$: $\dot{\tilde{\mathbf{x}}} = (A - LC)\tilde{\mathbf{x}}$.
**Condition**: $A{-}LC$ stable ($\Rightarrow \tilde{\mathbf{x}}{\to}0$). Roots (eig) of $A{-}LC$: **negative** $\Rightarrow$ error decays, $\hat{\mathbf{x}} \to \mathbf{x}$ (convergent); **positive** $\Rightarrow$ unstable, error grows, never converges.
Design: $\det(sI - (A - LC)) \overset{!}{=}$ desired poly, match coeffs.
***Output feedback:*** (separation): observer + $u = -K\hat{\mathbf{x}} + Nr$. CL char poly: $\det(sI{-}(A{-}BK))\det(sI{-}(A{-}LC))$.
***Separation principle:*** systems *both reachable & observable*: design

cont $K$ and obsv $L$ **independently**; the CL poles are just the union of the controller poles (eig $A-BK$) and observer poles (eig $A-LC$).
***Feedforward:*** (zero SS err): $N = -\left(C(A-BK)^{-1}B\right)^{-1}$.
***Integral action:*** augment state $\dot{q} = C\mathbf{x} - \bar{y}$ to kill SS error without exact model.

## 12. Frequency Response (Lec 9)

Input $A\sin\omega t \Rightarrow y_{ss} = A|G(j\omega)|\sin(\omega t + \phi)$
$G(j\omega) = |G(j\omega)|e^{j\phi}$, $\phi = \angle G = \arctan\frac{\text{Im}}{\text{Re}}$.
$\angle G = \sum\angle(\text{num}) - \sum\angle(\text{den})$: **numerator** (zeros) add +ve phase, **denominator** (poles) add −ve phase.
***Complex algebra:*** $z_1 z_2$: multiply mags, add phases; $z_1/z_2$: divide mags, subtract phases.
Ex. $G = \dfrac{1}{j\omega(j\omega+1)(j\omega+3)}$:

$$|G| = \frac{1}{\sqrt{\omega^2(\omega^2+1)(\omega^2+9)}}$$

$$\angle G = -\left(\tfrac{\pi}{2} + \arctan\omega + \arctan\tfrac{\omega}{3}\right)$$

***Forced sinusoid (undetermined coeffs):*** for $\ddot{x} + \omega_n^2 x = F\sin\omega t$, try $x{=}X\sin\omega t$, match:

$$X = \frac{F}{\omega_n^2 - \omega^2}$$

real ⇒ phase $0°/180°$ (no lag); $X \to \infty$ as $\omega \to \omega_n$ (**resonance**, undamped).

## 13. Bode Plots (Lec 9–10)

Mag in dB: $20\log_{10}|G(j\omega)|$. $K = 10^{dB/20}$, $dB = 20\log_{10}K$. $\times 10 \Rightarrow$ 20dB; $1 \to 0$dB. ***Standard form:*** factor so each bracket starts with 1; constant out front = DC gain $K$:

$$G(s) = \frac{K(\frac{s}{a}+1)}{s^n(\frac{s}{b}+1)}$$

***Start height:*** $H(j\omega)_{\text{dB}} = 20\log_{10}(K\omega) = 20\log_{10}(K) + 20\log_{10}(\omega)$
***Mag slopes:*** (added at break freq $\omega{=}|p|, |z|$):

| | |
|---|---|
| DC gain $K$ | flat at $20\log K$ |
| integrator $1/s^n$ | $-20n$ dB/dec from $\omega{\to}0$ |
| diff. $s$ | +20 dB/dec |
| pole $(\frac{s}{b}{+}1)^{-1}$ | −20 dB/dec after $\omega{=}b$ |
| zero $(\frac{s}{a}{+}1)$ | +20 dB/dec after $\omega{=}a$ |
| cplx poles @ $\omega_n$ | −40 dB/dec; peak $M_z = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$ at $\omega_n$ |
| cplx zeros @ $\omega_n$ | +40 dB/dec |

***Phase:*** $\angle G = \sum\angle(\text{num}) - \sum\angle(\text{den})$; each $(j\omega{+}a)$: $\arctan\frac{\omega}{a}$; transition ±1 decade about corner.

| | |
|---|---|
| DC gain | 0° |
| integrator $1/s$ | −90° (all $\omega$) |
| diff. $s$ | +90° (all $\omega$) |
| pole @ $b$ | −90° total, centred $b$ |
| zero @ $a$ | +90° total, centred $a$ |
| cplx poles @ $\omega_n$ | −180° total |

***Bode → TF:*** **(1)** DC gain $K{=}10^{X\text{dB}/20}$ from flat low-$\omega$ ($\omega = 1$ if slanted start). **(2)** initial slope: flat=no int., −20/dec=one $1/s$, $-40{=}1/s^2$, $+20{=}s$. **(3)** corners where slope *changes*: extra −20 ⇒ pole, +20 ⇒ zero, ∓40 ⇒ cplx pair @ $\omega_n$. **(4)** phase confirms: start 0/−90/−180°=0/1/2 integrators; drop @ corner=pole, rise=zero. **(5)** assemble:

$$G(s) = \frac{K\,s^{\text{pos}}(s{+}z_1)(s{+}z_2)\cdots}{s^{\text{int}}(s{+}p_1)(s{+}p_2)\cdots}$$

***Sanity:*** final slope = $-20(\#p{-}\#z)$ dB/dec; final phase = $-90°(\#p{-}\#z)$; DC check: sub $s{=}0$.

***GM & PM on Bode:*** **GM** = magnitude (dB) at phase = −180°; **PM** = phase (deg) at magnitude 0 dB.

## 14. Nyquist Criterion (Lec 10,12)

Plot $L(j\omega)$, $\omega: -\infty \to \infty$. Closed loop $G_{yr} = \frac{L}{1+L}$.
**Theorem**: $P$=open-loop RHP poles of $L$; $N$=net *clockwise* encirclements of −1. Then CL has $Z = N + P$ RHP poles.
**Stable** ⇔ $Z = 0$ ⇔ $N = -P$ ($P$ anticlockwise encirclements).
Stable-$L$ case: stable iff *no* encirclements of −1.
CL poles satisfy $L(s) = -1$.
***Real-axis crossing:*** Nyquist evaluates $G$ along the imag. axis, so to find where it crosses $(\alpha, 0)$: set $G(s){=}\alpha$ and solve — the root is **purely imaginary** $s{=}\pm j\omega$.

## 15. Stability Margins (Lec 10,11)

Crossover $\omega_{gc}$: $|L(j\omega_{gc})| = 1$ (0 dB).
***Gain margin:*** $g_m = 1/|L(j\omega_{pc})|$ at phase = −180°. *Find where Nyquist crosses −ve real axis; GM = inverse of that magnitude. (+ve).*
***Phase margin:*** $\varphi_m = 180° + \angle L(j\omega_{gc})$. *Find where Nyquist hits unit circle; PM = angle from that point to −ve x-axis.*
***Stability margin:*** $s_m$=closest dist of Nyquist to −1; $|S(j\omega)|$ = 1/dist to (−1). *Shortest distance from Nyquist curve to (−1, 0).* $|S(j\omega)|$ = sensitivity.
$PM = 2\sin^{-1}\left(\frac{1}{2|S(j\omega_c)|}\right)$.
***Time delay:*** Max delay $\tau_{max} = \varphi_m/\omega_{gc}$.

## 16. Sensitivity / Gang of Four (Lec 11)

$$S = \frac{1}{1+PC}, \qquad T = \frac{PC}{1+PC}, \qquad S + T = 1$$

$S{+}T{=}1$: a seesaw — low $S$ (tracking/dist. rejection) vs. low $T$ (robust to noise/model uncertainty); shape each small in different bands.
***3 TFs:*** $d, r, n$. → Set other 2 to 0 and solve.
Disturbance: $\eta = PS\,d$. $|S|{<}1$ attenuates; $|S|{=}0$ eliminates; $|S|{>}1$ worsens.
$S$ measures sensitivity of CL to plant uncertainty.
***Loop shaping — why:*** shape $L{=}PC$ to balance performance vs stability.
***how:*** (i) high gain / integrator at low $\omega$ → tracking + reject $d$ ($S{\approx}0$); (ii) low gain + steep roll-off at high $\omega$ → reject noise $n$ ($T{\approx}0$); (iii) gentle slope (≈ −20dB/dec) near $\omega_{gc}$ → keep phase margin.
Or pick desired $L$ (e.g. $\frac{\alpha}{s}$, $\alpha$ sets $\omega_{gc}$) then $C{=}L/P$ — avoid cancelling unstable poles/zeros.
***Robustness:*** relative model error $\Delta m(s){=}\dfrac{\Delta G(s)}{G(s)}$; robust stability $\Leftarrow |T(j\omega)| < \dfrac{1}{|\Delta m(j\omega)|}$ — keep $T$ small where $\Delta m$ large (high $\omega$).

## 17. Fundamental Limitations (Lec 11,12)

**(1)** $S + T = 1$ — seesaw trade-off.
**(2)** Gain–phase relation — steep roll-off ⇒ phase lag, low PM.
**(3)** Interpolation constraints — RHP poles/zeros can't be hidden.
**(4)** Bode sensitivity integral (waterbed):

$$\int_0^\infty \log|S(j\omega)|\,d\omega = \pi\sum_k p_k$$

$p_k$=unstable open-loop poles; push $|S|$ down somewhere ⇒ up elsewhere.

## 19. Block Diagram Algebra

***Series:*** $G_1G_2$. ***Parallel:*** $G_1 \pm G_2$.
***Neg. feedback:*** $\dfrac{G}{1+GH}$; unity: $\dfrac{G}{1+G}$.
***Pos. feedback:*** $\dfrac{G}{1-GH}$.

$L(s) = C(s)P(s)$ (loop TF), $T(s) = \frac{L(s)}{1+L(s)}$, $S(s) = \frac{1}{1+L(s)}$ (sensitivity function maps disturbance on plant output to final output).
***Cascaded SS — recipe:*** EOM of each subsystem:

$$\dot{x}_1{=}A_1x_1{+}B_1u_1,\ y_1{=}C_1x_1; \quad \dot{x}_2{=}A_2x_2{+}B_2u_2,\ y_2{=}C_2x_2$$

Sub interconnection $u_2{=}y_1{=}C_1x_1$ into $\dot{x}_2$; stack $x{=}[x_1; x_2]$:

$$\dot{x} = \begin{bmatrix} A_1 & 0 \\ B_2C_1 & A_2 \end{bmatrix} x + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u_1, \quad y_2 = \begin{bmatrix} 0 & C_2 \end{bmatrix} x$$

***TF:*** get $G_i{=}C_i(sI{-}A_i)^{-1}B_i{+}D_i$ for each subsystem, then $G_{tot}{=}G_2G_1$ (multiply) — far easier than inverting the big $(sI{-}A)$.

## 20. Quick Reference / Exam Tips

- Stable ⇔ all poles/eigenvalues in open LHP.
- Reachable: $\det[B\ AB] \neq 0$; Observable: $\det[C; CA] \neq 0$ (2nd order).
- PD design: pick $\zeta$ from $M_p$, $\omega_n$ from $t_s$, match denom.
- Nyquist through $(\alpha, 0)$: set $G(s) = \alpha$; root is purely imaginary $s{=}\pm j\omega$.
- Zero SS error tracking: use **integral** action (robust) or feedforward $N$ (needs exact model).
- Real-world uncertainty: parameter errors, disturbances, sensor noise, unmodelled nonlinearity.

***PID vs state-feedback:*** **PID**: model-free, mainly SISO, heuristic/robust, simple to tune, limited pole control. **State-FB**: needs accurate model + full state (or observer), handles MIMO, gives exact pole placement.
***Useful:*** $\zeta{=}0.5{\Rightarrow}M_p{\approx}16\%$; $\zeta{=}0.707{\Rightarrow}M_p{\approx}4.3\%$; $\zeta{=}0.6{\Rightarrow}M_p{\approx}9.5\%$. $\sqrt{1-\zeta^2}$ at $\zeta{=}0.707$ is 0.707.

## 21. Written Question Framework

***Observability:*** Can uniquely recover $\mathbf{x}(0)$ from measured $u, y$ over $[0, T]$. Find unmeausrable from what you can measure.
***Observer purpose:*** estimate unmeasured state $\mathbf{x}$ from available $u, y$ → enables state-feedback when full state unavailable.
***Sensitivity function $S$:*** measures system sensitivity to plant uncertainty; $|S| \ll 1$ means disturbances attenuated (good tracking, disturbance rejection). $S{+}T = 1$ (seesaw): cannot make both small.
***$S$ & $T$ trade-off:*** $S$ low at low $\omega$ (track, reject slow disturbances); $T$ low at high $\omega$ (reject fast noise). Near crossover $\omega_{gc}$, phase margin enforces balance.
***Gain margin:*** $GM = 1/|L(j\omega_{pc})|$ where phase = −180°. On Nyquist, find where curve crosses negative real axis; GM = reciprocal of magnitude. On Bode, read magnitude (dB) at phase −180°.
***Real-world control design:*** (1) model system (ODE/SS from physics); (2) ID parameters (expts, data); (3) check reachable/observable; (4) design $K$ for transient specs $(\zeta, \omega_n)$; (5) design $L$ for measurement noise tolerance; (6) check closed-loop stability (Nyquist/poles); (7) simulate with real noise/uncertainty.
***Step response features from TF:*** $G(s)$ poles ⇒ poles in TF ⇒ envelope decay $e^{-\sigma t}$ and oscillation frequency $\omega_d$. Final value $x(\infty) = G(0)\cdot U$. DC gain sign ⇒ direction (dip/rise).

## 22. Math Toolbox

$2\times 2$ ***inverse:*** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

$3\times 3$ ***det:*** $\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei{-}fh) - b(di{-}fg) + c(dh{-}eg)$.

$2\times 2$ ***eigenvalues:*** $\det(\lambda I - A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$

***Common integrals:*** $\int e^{-at}dt = -\frac{1}{a}e^{-at}$; $\int_0^\infty e^{-at}dt = \frac{1}{a}$ $(a{>}0)$.

***Sin/cos deriv:*** $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$, $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$.

***Sin/cos integ:*** $\int \sin\omega t\,dt = -\frac{1}{\omega}\cos\omega t$, $\int\cos\omega t\,dt = \frac{1}{\omega}\sin\omega t$.